

SMITH FIELD, AR, USA

WMO: 723443

Lat: 36.191N

Lon: 94.491W

Elev: 364

StdP: 97.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53955

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.8	-17.5	0.8	-10.5	-15.0	1.1	-7.3	11.9	9.3	11.0	9.9	3.3	320	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	36.1	23.7	33.9	23.7	32.5	23.6	25.4	32.3	24.8	31.6	24.2	30.8	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.6	29.1	22.7	18.2	28.6	22.3	17.8	28.3	79.7	32.1	77.0	31.8	75.0	31.0	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.1	8.2	DB	-15.4	37.6	2.9	2.7	-17.5	39.5	-19.2	41.1	-20.9	42.6	-23.0	44.6	(4)
(5)			WB	-16.2	26.3	2.7	0.8	-18.1	26.9	-19.7	27.4	-21.2	27.9	-23.2	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	2.7	5.0	9.5	14.7	19.3	23.5	26.2	25.8	21.3	15.4	9.4	4.1	(6)
(7)		DBStd	9.49	6.27	6.28	5.89	4.92	4.01	2.97	2.73	2.93	3.99	4.83	5.26	5.66	(7)
(8)		HDD10.0	777	238	162	80	16	1	0	0	0	11	74	196		(8)
(9)		HDD18.3	2149	485	374	276	127	39	3	0	1	18	116	271	441	(9)
(10)		CDD10.0	2528	12	22	66	156	288	406	503	489	339	178	55	14	(10)
(11)		CDD18.3	857	0	1	3	18	68	158	245	232	106	24	2	0	(11)
(12)		CDH23.3	8155	0	3	19	119	497	1436	2604	2360	911	196	10	0	(12)
(13)		CDH26.7	3425	0	1	1	17	116	530	1248	1138	333	40	1	0	(13)
(14)	Wind	WSAvg	3.5	3.9	4.1	4.3	4.5	3.6	3.2	2.8	2.7	2.8	3.2	3.8	3.8	(14)
(15)	Precipitation	PrecAvg	1169	65	58	93	118	151	116	97	76	108	98	97	79	(15)
(16)		PrecMax	1703	173	157	236	316	292	312	271	145	268	252	247	317	(16)
(17)		PrecMin	826	4	9	18	45	57	28	27	7	11	32	13	10	(17)
(18)		PrecStd	218	45	40	49	63	56	65	56	38	57	48	71	59	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	22.5	25.9	28.7	31.3	35.1	37.9	38.0	35.8	30.0	24.9	20.6	(19)
(20)			MCWB	14.1	15.2	17.1	19.1	22.1	24.2	23.2	23.6	22.7	21.0	16.8	15.7	(20)
(21)		2%	DB	17.5	19.1	22.9	26.3	29.1	32.7	36.1	36.2	32.3	27.6	22.2	17.8	(21)
(22)			MCWB	12.7	13.5	15.5	17.8	21.6	23.8	24.0	23.4	22.2	19.0	15.5	13.7	(22)
(23)		5%	DB	15.1	17.4	21.1	24.1	27.7	31.3	33.9	33.9	30.8	25.9	19.9	16.0	(23)
(24)			MCWB	11.0	12.2	14.2	16.7	20.9	23.4	24.1	23.5	21.9	18.0	15.0	11.8	(24)
(25)		10%	DB	12.6	15.2	18.8	22.6	26.3	29.8	32.5	32.4	28.0	23.0	17.8	12.9	(25)
(26)			MCWB	8.5	10.5	13.3	15.9	19.9	22.8	24.1	23.4	20.7	17.1	13.2	9.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	16.9	18.4	20.5	23.8	25.7	26.3	26.2	24.4	22.0	18.8	16.8	(27)
(28)			MCDWB	19.0	20.9	24.1	26.3	28.9	32.2	32.1	33.1	30.9	27.6	22.6	19.7	(28)
(29)		2%	WB	13.7	15.0	16.7	19.2	22.8	24.8	25.4	25.3	23.7	20.6	17.0	14.6	(29)
(30)			MCDWB	16.5	18.2	21.7	24.6	27.9	31.0	32.3	32.4	29.8	25.8	20.4	17.5	(30)
(31)		5%	WB	11.4	13.2	15.3	18.1	21.7	24.0	24.9	24.6	22.8	19.3	15.5	12.4	(31)
(32)			MCDWB	14.4	16.6	19.4	22.8	26.5	29.6	31.8	31.6	28.7	23.8	18.9	15.1	(32)
(33)		10%	WB	8.6	10.9	13.9	17.1	20.6	23.3	24.4	23.9	21.8	18.0	13.9	10.1	(33)
(34)			MCDWB	12.4	14.7	18.2	21.7	25.4	28.8	31.1	30.8	26.9	22.4	17.2	12.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	10.8	11.1	11.5	10.6	10.5	11.1	11.4	11.2	12.1	10.9	9.9	(35)
(36)			MCDBR	13.3	13.3	13.1	12.7	11.5	12.0	12.9	13.8	13.1	13.5	12.8	13.1	(36)
(37)			MCWBR	9.4	9.5	8.3	7.6	6.3	5.6	4.9	5.0	5.8	7.3	8.6	9.5	(37)
(38)		5% WB	MCDWR	11.2	11.6	11.1	10.7	10.2	10.6	11.4	11.4	10.7	11.1	11.3	11.1	(38)
(39)			MCWBR	9.1	9.3	8.0	7.4	6.1	5.7	5.1	4.9	5.4	6.8	8.9	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.306	0.342	0.379	0.422	0.424	0.443	0.444	0.388	0.345	0.313	0.292	(40)	
(41)		taud	2.486	2.488	2.418	2.323	2.256	2.277	2.274	2.247	2.372	2.479	2.520	2.533	(41)	
(42)		Ebn at Noon	898	938	926	903	865	860	840	832	865	880	879	890	(42)	
(43)		Edh at Noon	82	92	109	126	137	134	134	135	113	92	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.46	2.99	4.03	5.07	5.62	6.43	6.41	5.81	4.95	3.68	2.65	2.10	(44)	
(45)		RadStd	0.22	0.43	0.24	0.29	0.48	0.59	0.39	0.39	0.37	0.48	0.30	0.27	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page